Tutorial – 1

1. Write a program to print “Hello World”.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_1
{
    internal class Q1
    {
        static void Main()
        {
            Console.WriteLine("Hello World");
        }
    }
}
```

Output:

```
D:\RKU\5th SEM\.NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial_1>Q1
Hello World
```

2. Design your profile page as given below.

```
namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_1
{
    internal class Q2
    {
        static void Main()
        {
            Console.WriteLine("Name: Saurav Pandit");
            Console.WriteLine("DOB: 20/11/2005");
            Console.WriteLine("Address: 8, Manda Dungar, Aji Dam");
            Console.WriteLine("City: Rajkot");
```

```
            Console.WriteLine("Pincode: 360 003");
            Console.WriteLine("State: Gujarat");
            Console.WriteLine("Country: India");
            Console.WriteLine("Email: srv@gmail.com");
        }
    }
}
```

Output:

```
D:\RKU\5th SEM\.NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial_1>Q2
Name: Saurav Pandit
DOB: 20/11/2005
Address: 8, Manda Dungar, Aji Dam
City: Rajkot
Pincode: 360 003
State: Gujarat
Country: India
Email: srv@gmail.com
```

3. Find out whether the given number is odd or even.

```
namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_1
{
    internal class Q3
    {
        static void Main()
        {
            Console.Write("Enter a number: ");
            int number = Convert.ToInt32(Console.ReadLine());

            if (number % 2 == 0)
            {
                Console.WriteLine(number + " is Even.");
            }
            else
            {
                Console.WriteLine(number + " is Odd.");
            }
        }
    }
}
```

Output:

```
D:\RKU\5th SEM\.NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial_1>Q3
Enter a number: 25
25 is Odd.
```

4. Rearrange the given code to correct the program. The resultant program will be to input a number and print whether the given number is odd or even.

```
namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_1
{
   internal class Q4
   {
      static void Main(string[] args)
      {
         int x;
         string str;

         Console.WriteLine("Enter Number: ");
         str = Console.ReadLine();

         x = Convert.ToInt32(str);

         if (x % 2 == 0)
            Console.WriteLine("Number is Even");
         else
            Console.WriteLine("Number is Odd");

         Console.Read();
      }
   }
}
```

Output:

```
D:\RKU\5th SEM\.NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial_1>Q4
Enter Number:
10
Number is Even
```

5. Write output of the program. Also write comment for each line for the following code.

```
namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_1
{
    internal class Q5
    {
        static void Main(string[] args)
        {
            int n, fact = 1;

            Console.WriteLine("Enter Number : ");

            string str = Console.ReadLine();

            n = Convert.ToInt32(str);

            for (int i = 1; i <= n; i++)
            {
                fact = fact * i;
            }

            Console.WriteLine("Factorial : {0}", fact);

            Console.Read();
        }
    }
}
```

Output:

```
D:\RKU\5th SEM\.NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial_1>Q5
Enter Number :
5
Factorial : 120
```

6. Write missing statement to get the desired output.

```
namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_1
{
    internal class Q6
```

```
    {
        static void Main(string[] args)
        {
            int a, b, c, result;
            string str;

            Console.Write("Enter Number   1: ");
            str = Console.ReadLine();
            a = Convert.ToInt32(str);

            Console.Write("Enter Number  2 : ");
            str = Console.ReadLine();
            b = Convert.ToInt32(str);

            Console.Write("Enter Number   3 : ");
            str = Console.ReadLine();
            c = Convert.ToInt32(str);

            result = Sum(a, b, c);
            Console.WriteLine("Sum : " + result);

            Console.Read();
        }

        static int Sum(int x, int y, int z)
        {
            int res;
            res = x + y + z;
            return res;
        }
    }
}
```

Output:

```
D:\RKU\5th SEM\.NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial_1>Q6
Enter Number   1: 10
Enter Number  2 : 20
Enter Number   3 : 30
Sum : 60
```

7. Predict and write the output of the given code.

Output:

```
Enter a number
5

Enter a number
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

8. Write a program to convert given name in upper characters.

```
namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_1
{
    internal class Q8
    {
        static void Main(string[] args)
        {
            string name;

            Console.Write("Enter your name: ");
            name = Console.ReadLine();

            string upperName = name.ToUpper();

            Console.WriteLine("Output: " + upperName);

            Console.ReadLine();
        }
    }
}
```

Output:

```
D:\RKU\5th SEM\.NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial_1>Q8
Enter your name: John F Kennady
Output: JOHN F KENNADY
```

9. Write a Program to convert given name in toggle case.

```
namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_1
{
    internal class Q9
    {
        static void Main(string[] args)
        {
            string name, toggleName = "";

            Console.Write("Enter your name: ");
            name = Console.ReadLine();

            foreach (char ch in name)
            {
                if (char.IsUpper(ch))
                    toggleName += char.ToLower(ch);
                else if (char.IsLower(ch))
                    toggleName += char.ToUpper(ch);
                else
                    toggleName += ch; // For spaces or symbols
            }

            Console.WriteLine("Output: " + toggleName);

            Console.ReadLine();
        }
    }
}
```

Output:

```
D:\RKU\5th SEM\.NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial_1>Q9
Enter your name: JoHn f KeNneDY
Output: jOhN F kEnNEdy
```

10. Write a Program which accepts mobile no as a string from the user and converts the last 5 digits into X.

```
namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_1
{
    internal class Q10
    {
        static void Main(string[] args)
        {
            string mobile;

            Console.Write("Enter your mobile number: ");
            mobile = Console.ReadLine();

            if (mobile.Length >= 5)
            {
                string masked = mobile.Substring(0, mobile.Length - 5) + "XXXXX";
                Console.WriteLine("Output: " + masked);
            }
            else
            {
                Console.WriteLine("Invalid mobile number. Must be at least 5 digits.");
            }

            Console.ReadLine();
        }
    }
}
```

Output:

```
D:\RKU\5th SEM\.NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial_1>Q10
Enter your mobile number: 1234567890
Output: 12345XXXXX
```

11. Write a Program which accepts name and gender from the user. Here, gender may have only 1 character, M or F.

```
namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_1
{
```

```
    internal class Q11
    {
        static void Main(string[] args)
        {
            string name;
            char gender;

            Console.Write("Enter your name: ");
            name = Console.ReadLine();

            Console.Write("Enter your gender (M/F): ");
            gender = Convert.ToChar(Console.ReadLine().ToUpper());

            if (gender == 'M')
            {
                Console.WriteLine("Output: Mr. " + name);
            }
            else if (gender == 'F')
            {
                Console.WriteLine("Output: Ms. " + name);
            }
            else
            {
                Console.WriteLine("Invalid gender input. Please enter M or F.");
            }

            Console.ReadLine();
        }
    }
}
```

Output:

```
D:\RKU\5th SEM\.NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial_1>Q11
Enter your name: Hillary Clinton
Enter your gender (M/F): F
Output: Ms. Hillary Clinton
```

12. Write a Program which accepts name from the user and prints the same

```
namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_1
```

```
{
    internal class Q12
    {
        static void Main(string[] args)
        {
            string name;

            Console.Write("Enter your name: ");
            name = Console.ReadLine();

            Console.WriteLine("Output: " + name);

            Console.ReadLine();
        }
    }
}
```

Output:

```
D:\RKU\5th SEM\.NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial_1>Q12
Enter your name: Saurav Pandit
Output: Saurav Pandit
```

13. Write a Program to prints the following series
0 1 1 2 3 5 8 13 21 34 55

```
namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_1
{
    internal class Q13
    {
        static void Main(string[] args)
        {
            int a = 0, b = 1, c;

            Console.Write(a + " " + b + " ");

            while (true)
            {
                c = a + b;
                if (c > 55)
                    break;
```

```
                Console.Write(c + " ");
                a = b;
                b = c;
            }

            Console.ReadLine();
        }
    }
}
```

Output:

```
D:\RKU\5th SEM\.NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial_1>Q13
0 1 1 2 3 5 8 13 21 34 55
```

14. Write a Program which accepts no from the user and print the same in words.

```
namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_1
{
    internal class Q14
    {
        static void Main(string[] args)
        {
            Console.Write("Enter a number: ");
            string number = Console.ReadLine();

            foreach (char digit in number)
            {
                switch (digit)
                {
                    case '0': Console.Write("Zero "); break;
                    case '1': Console.Write("One "); break;
                    case '2': Console.Write("Two "); break;
                    case '3': Console.Write("Three "); break;
                    case '4': Console.Write("Four "); break;
                    case '5': Console.Write("Five "); break;
                    case '6': Console.Write("Six "); break;
                    case '7': Console.Write("Seven "); break;
                    case '8': Console.Write("Eight "); break;
```

```
                case '9': Console.Write("Nine "); break;
                default: Console.Write("? "); break;
            }
        }

        Console.ReadLine();
    }
  }
}
```

Output:

```
D:\RKU\5th SEM\.NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial_1>Q14
Enter a number: 98732
Nine Eight Seven Three Two
```

15. Write a Program to check whether the given no is Armstrong no or not.

```
namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_1
{
  internal class Q15
  {
    static void Main(string[] args)
    {
      int number, originalNumber, remainder, result = 0, digits = 0;

      Console.Write("Enter a number: ");
      number = Convert.ToInt32(Console.ReadLine());
      originalNumber = number;

      int temp = number;
      while (temp != 0)
      {
        temp /= 10;
        digits++;
      }

      temp = number;
      while (temp != 0)
      {
        remainder = temp % 10;
        result += (int)Math.Pow(remainder, digits);
        temp /= 10;
```

```
            }

            if (result == originalNumber)
                Console.WriteLine(originalNumber + " is an Armstrong number.");
            else
                Console.WriteLine(originalNumber + " is not an Armstrong number.");

            Console.ReadLine();
        }
    }
}
```

Output:

```
D:\RKU\5th SEM\.NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial_1>Q15
Enter a number: 153
153 is an Armstrong number.


D:\RKU\5th SEM\.NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial_1>csc Q15.cs
Microsoft (R) Visual C# Compiler version 4.14.0-3.25279.5 (995f12b6)
Copyright (C) Microsoft Corporation. All rights reserved.


D:\RKU\5th SEM\.NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial_1>Q15
Enter a number: 286
286 is not an Armstrong number.
```

16. Write a program to display a pattern like a right angle triangle using an asterisk.

```
namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_1
{
    internal class Q16
    {
        static void Main(string[] args)
        {
            for (int i = 1; i <= 4; i++)
            {
                for (int j = 1; j <= i; j++)
                {
                    Console.Write("*");
                }
                Console.WriteLine();
            }
```

```
            Console.ReadLine();
        }
    }
}
```

Output:

```
D:\RKU\5th SEM\.NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial_1>Q16
*
**
***
****
```

17. Write a Program to generate following output.

```
namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_1
{
    internal class Q17
    {
        static void Main(string[] args)
        {
            for (int i = 1; i <= 4; i++)
            {
                for (int j = 1; j <= i; j++)
                {
                    Console.Write(j + " ");
                }
                Console.WriteLine();
            }

            Console.ReadLine();
        }
    }
}
```

Output:

```
D:\RKU\5th SEM\.NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial_1>Q17
1
1 2
1 2 3
1 2 3 4
```

18. Write a program to make such a pattern like a right angle triangle with the number increased by 1.

```
namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_1
{
    internal class Q18
    {
        static void Main(string[] args)
        {
            int num = 1;

            for (int i = 1; i <= 4; i++)
            {
                for (int j = 1; j <= i; j++)
                {
                    Console.Write(num + " ");
                    num++;
                }
                Console.WriteLine();
            }

            Console.ReadLine();
        }
    }
}
```

Output:

```
D:\RKU\5th SEM\.NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial_1>Q18
1
2 3
4 5 6
7 8 9 10
```

19. Write a program to make such a pattern as a pyramid with an asterisk.

```
namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_1
{
    internal class Q19
    {
        static void Main(string[] args)
        {
            int rows = 4;
```

```
            for (int i = 1; i <= rows; i++)
            {
                for (int space = 1; space <= rows - i; space++)
                {
                    Console.Write(" ");
                }

                for (int star = 1; star <= i; star++)
                {
                    Console.Write("* ");
                }

                Console.WriteLine();
            }

            Console.ReadLine();
        }
    }
}
```

Output:

```
D:\RKU\5th SEM\.NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial_1>Q19
   *
  * *
 * * *
* * * *
```

20. Write a program to make a pyramid pattern with numbers increased by 1.

```
namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_1
{
    internal class Q20
    {
        static void Main(string[] args)
        {
            int rows = 4;
            int num = 1;

            for (int i = 1; i <= rows; i++)
            {
                for (int space = 1; space <= rows - i; space++)
```

```
                {
                    Console.Write(" ");
                }

                for (int j = 1; j <= i; j++)
                {
                    Console.Write(num + " ");
                    num++;
                }

                Console.WriteLine();
            }

            Console.ReadLine();
        }
    }
}
```

Output:

```
D:\RKU\5th SEM\.NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial_1>Q20
   1
  2 3
 4 5 6
7 8 9 10
```

21. Write a program to find the sum of the series 5 +55 + 555 + 5555 + .. n terms.

```
namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_1
{
    internal class Q21
    {
        static void Main(string[] args)
        {
            int n, digit;

            Console.Write("Input the number of terms: ");
            n = Convert.ToInt32(Console.ReadLine());

            Console.Write("Input number: ");
            digit = Convert.ToInt32(Console.ReadLine());

            int term = 0;
```

```
            int sum = 0;

            Console.Write("Series: ");

            for (int i = 1; i <= n; i++)
            {
                term = term * 10 + digit;
                sum += term;

                Console.Write(term);
                if (i != n)
                    Console.Write(" + ");
            }

            Console.WriteLine();
            Console.WriteLine("The Sum is : " + sum);

            Console.ReadLine();
        }
    }
}
```

Output:

```
D:\RKU\5th SEM\.NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial_1>Q21
Input the number of terms: 4
Input number: 5
Series: 5 + 55 + 555 + 5555
The Sum is : 6170
```

22. Write a program to display a pattern like a diamond.

```
namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_1
{
    internal class Q22
    {
        static void Main(string[] args)
        {
            int n = 5;

            for (int i = 1; i <= n; i++)
            {
                for (int space = 1; space <= n - i; space++)
```

```
                {
                    Console.Write(" ");
                }

                for (int star = 1; star <= 2 * i - 1; star++)
                {
                    Console.Write("*");
                }

                Console.WriteLine();
            }

            for (int i = n - 1; i >= 1; i--)
            {
                for (int space = 1; space <= n - i; space++)
                {
                    Console.Write(" ");
                }

                for (int star = 1; star <= 2 * i - 1; star++)
                {
                    Console.Write("*");
                }

                Console.WriteLine();
            }

            Console.ReadLine();
        }
    }
}
```

Output:

```
D:\RKU\5th SEM\.NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial_1>Q22
    *
   ***
  *****
 *******
*********
 *******
  *****
   ***
    *
```